



**AL RYADA UNIVERSITY FOR  
SCIENCE AND TECHNOLOGY**



# Medications administration



**By**

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**Lecture medical surgical nursing**

# Outlines

- Definition.
- Legal aspects of drug administration.
- Patients' rights related to medication administration.
- Ten “Rights” of medication administration.
- Effects of drugs.
- Actions of drugs on the body.
- Pharmacodynamics.

# Outlines

- Pharmacokinetics.
- Factors affecting medication action.
- Routes of drug administration.
- Types of medication orders.
- Essential parts of a drug order.
- Systems of medication measurement.

# Definition

**A medication** is a substance administered for the

Diagnosis,

Treatment,

relief of a symptom

or prevention of disease

# Definition

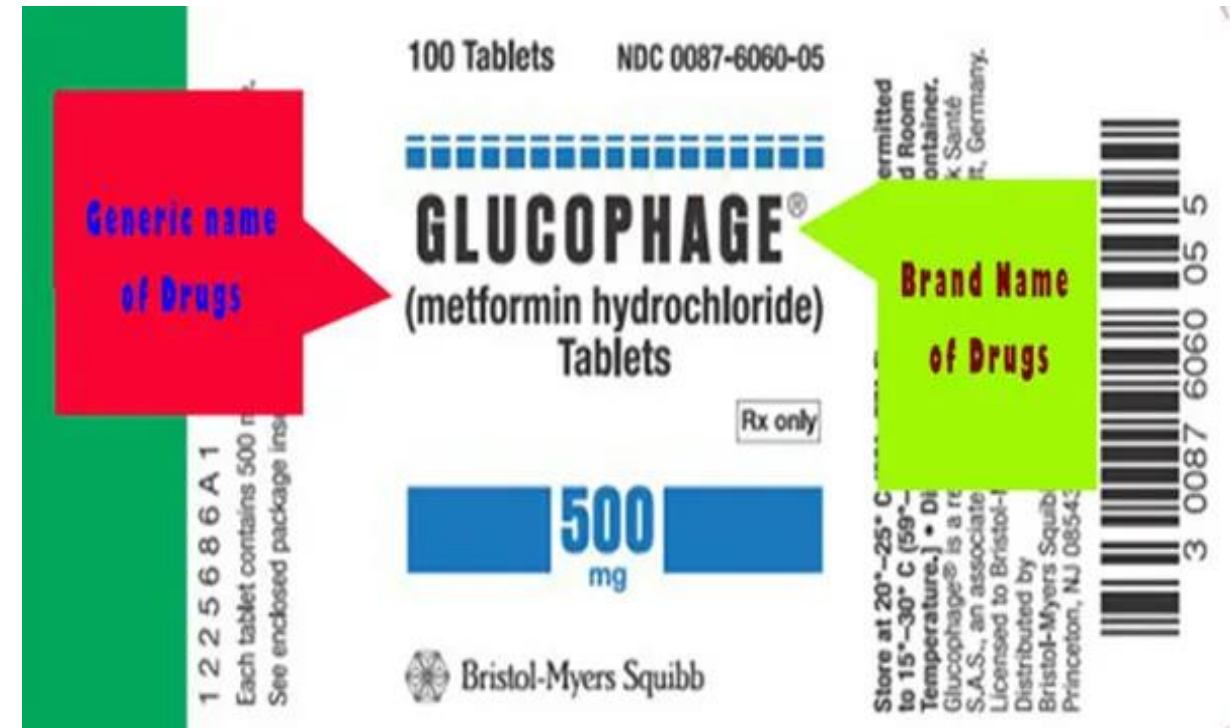
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- A **prescription** means the written direction for the preparation and administration of a drug.



- **Each drug can have four names:**

- Generic name
- Official name
- Chemical name
- Trademark or brand name.



# Definition

- **Generic name:** is assigned by the United States Adopted Names (USAN) Council and is used throughout the drug's lifetime. الاسم العلمى (المادة الفعالة)
- **Official name:** is the name under which it is listed in one of the official publication (United States pharmacopeia). الاسم الرسمى للدواء وغالبا الاسم العلمى

# Definition

- **Chemical name:** is the name by which a chemist knows it; this name describes the constituents of the drug.
- **Trade name (brand name):** is the name given by the drug manufacturer and identifies it as property of that company (هو الاسم الذي تعطيه شركة تصنيع الدواء للمنتج)



# Legal aspects of drug administration

Under the law, nurses are responsible for their own actions regardless of whether there is a written order. **المرضة تبقى مسؤولة قانونيًا عن أفعالها**

Another aspect of nursing practice governed by law is the use of controlled substances according to the agency policy in keeping, using and recording of this substance. **القانون ينظم طريقة تعامل الممرضة مع الأدوية المخدرة أو الخاضعة للرقابة (مثل المورفين وغيره)، ولازم تلتزم بسياسة المستشفى**

## Purpose of Patients' Rights related to medication administration

1. To be informed of the name, purpose, action, and potential undesired effects of a medication

2. To refuse a medication regardless of the consequences

3. To assess a medication history, including allergies and use of herbals

4. To be properly advised of the experimental nature of medication therapy and give written consent for its use



# Cont, Purpose of Patients' Rights related to medication administration

5. To receive labeled medications safely in accordance with the fourteen rights of medication administration

6. To receive appropriate supportive therapy in relation to medication therapy

7. To not receive unnecessary medications

8. To be informed if medications are a part of a research study

# fourteen “Rights” of Medication Administration

| Medication right         | Nursing intervention                                                                                                                                                                                                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Right medication name | - The medication given was the medication ordered                                                                                                                                                                                                                                                           |
| 2. Right dose            | <ul style="list-style-type: none"><li>- The dose ordered is appropriate for the client.</li><li>- Double-check calculations that appear questionable or indicate multiple pills/ tablets or a large quantity of a liquid medication.</li><li>- Question a dose outside of the usual dosage range.</li></ul> |
| 3. Right time            | - Give the medication at the right frequency and at the time ordered and guidelines according to agency policy.                                                                                                                                                                                             |
| 4. Right route           | - Give the medication by the ordered route and make sure that the route is safe and appropriate for the client.                                                                                                                                                                                             |
| 5. Right client          | - Medication is given to the intended client by checking the client's identification band with each administration of a medication.                                                                                                                                                                         |

# fourteen “Rights” of Medication Administration

| Medication right  | Nursing intervention                                                                                                                                                                                             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6- Right Reason   | <ul style="list-style-type: none"><li>-Confirm the medication is being given for the correct indication.</li><li>- Ensure the patient’s condition warrants the use of the medication.</li></ul>                  |
| 7- Right Response | Monitor the patient for the expected therapeutic effect.- Assess for any adverse reactions or side effects.                                                                                                      |
| 8-Right Storage   | <ul style="list-style-type: none"><li>-Ensure medications are stored properly to maintain their efficacy and safety.</li><li>- Follow guidelines for temperature, light, and humidity.</li></ul>                 |
| 9- Right Disposal | <ul style="list-style-type: none"><li>-Dispose of unused or expired medications safely and in accordance with regulations.</li><li>- Educate patients on proper disposal methods for home medications.</li></ul> |

## fourteen “Rights” of Medication Administration

|                           |                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10-Right client education | <ul style="list-style-type: none"><li>- Explain information about the medication to the client (e.g., why receiving, what to expect, any precautions).</li></ul>                                                                                                                                                                         |
| 11- Right documentation   | <ul style="list-style-type: none"><li>- Document medication administration after giving it, not before. If a medication is not given, follow the agency’s policy for documenting the reason why.</li></ul>                                                                                                                               |
| 12-Right to refuse        | <ul style="list-style-type: none"><li>- Adult clients have the right to refuse any medication, the nurse’s role is to ensure that the client is fully informed of the potential consequences of refusal and to communicate the client’s refusal to the health care provider</li></ul>                                                    |
| 13- Right assessment      | <ul style="list-style-type: none"><li>- Some medications require specific assessments prior to administration as apical pulse, blood pressure, lab results.</li><li>- Medication orders may include specific parameters for administration (e.g., do not give if pulse less than 60 or systolic blood pressure less than 100).</li></ul> |
| 14- Right evaluation      | <ul style="list-style-type: none"><li>- Conduct appropriate follow-up (e.g., was the desired effect achieved or not? Did the client experience any side effects or adverse reactions?).</li></ul>                                                                                                                                        |



# Effects of drugs

1. **Therapeutic effect(desired effect):** is the primary effect intended, that is, the reason the drug is prescribed.



## Table (2): Therapeutic Actions of Drugs

| Drug Type                  | Description                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>1. Palliative</b>       | Relieves the symptoms of a disease but does not affect the disease itself, e.g. Morphine sulfate for pain.                  |
| <b>2. Curative</b>         | Cures a disease or condition, Penicillin for infection.                                                                     |
| <b>3. Supportive</b>       | Supports body function until other treatments or the body's response can take over, e.g. aspirin for high body temperature. |
| <b>4. Substitutive</b>     | Replaces body fluids or substances, e.g. insulin for diabetes mellitus.                                                     |
| <b>5. Chemotherapeutic</b> | Destroys malignant cells                                                                                                    |
| <b>6. Restorative</b>      | Returns the body to health, e.g. Vitamin, mineral supplements.                                                              |



# Cont, Effects of drugs

۲. **A side effect (secondary effects):** is one that is unintended, are usually predictable and may be either harmless or potentially harmful. For example, digitalis increases the strength of myocardial contractions (desired effect), but it can have the side effect of inducing nausea and vomiting.

**Adverse effects or reactions:** are more severe side effects may justify the discontinuation of a drug. The nurse should monitor for dose-related side or adverse effects and report these to the health care provider who may discontinue the medication or change the dosage.

# Cont, Effects of drugs

**3. Drug allergy** is an immunologic reaction to a drug. When a client is first exposed to a foreign substance (antigen), the body may react by producing antibodies. A client may react to a drug as to an antigen and thus develop symptoms of an allergic reaction. Allergic reactions can be mild or severe.



# Drug allergy:



**A mild reaction:** has a variety of symptoms includes skin rash, pruritus, angioedema, lacrimal tearing, nausea, vomiting, wheezing, dyspnea and diarrhea.



**Severe allergic reaction:** usually occurs immediately after the administration of drug and is called an anaphylactic reaction. It can be fatal if symptoms are not noticed and treated immediately. The earliest symptoms are a subjective feeling of swelling in the mouth and tongue, acute shortness of breathing, acute hypotension, and tachycardia.

# Cont, Effects of drugs

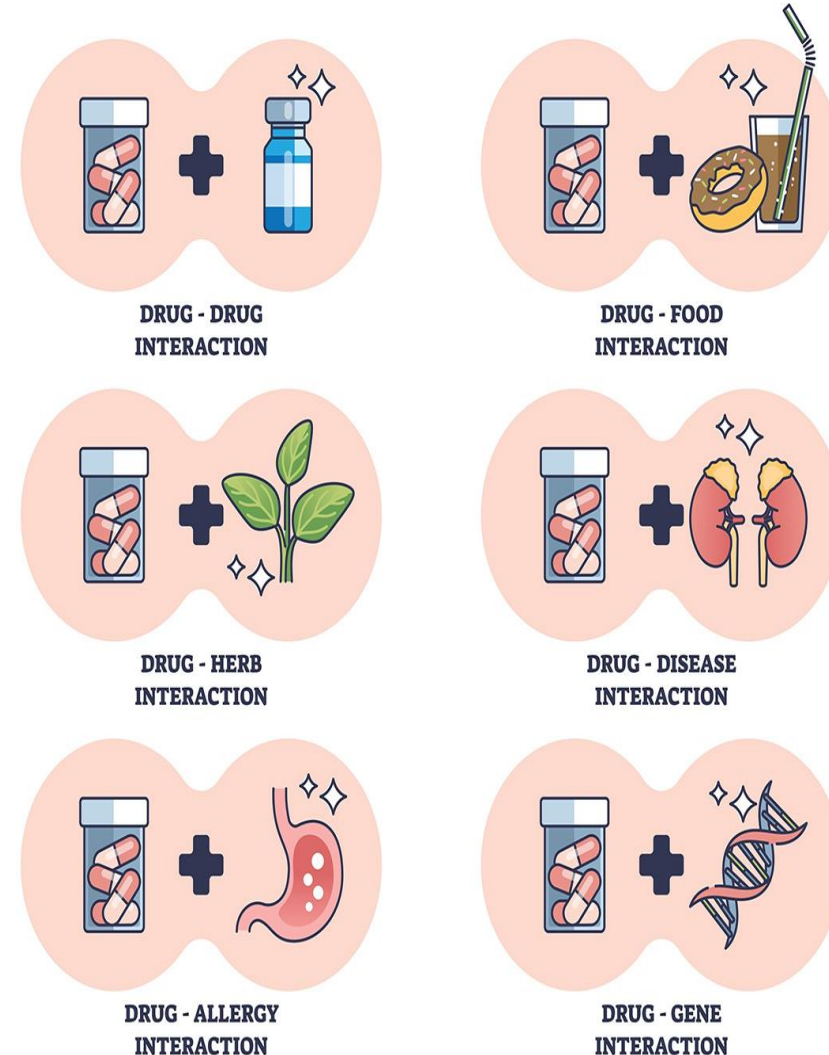
**4. Drug tolerance** انخفاض استجابة الجسم للدواء: exists in a person who exhibits an unusually low physiological response to a drug and who requires increases in the dosage to maintain a given therapeutic effect. Drugs that commonly produce tolerance are opiates.

**5. A cumulative effect** التأثير التراكمي: is the increasing response to repeated doses of a drug that occurs when the rate of administration exceeds the rate of metabolism or excretion. As a result, the amount of the drug builds up in the client's body unless the dosage is adjusted. Toxic symptoms may occur.

# Cont, Effects of drugs

6. A drug interaction occurs when the administration of one drug before, at the same time as, or after another drug **alters the effect of one or both drugs**. Drug interactions may be beneficial or harmful. The effect of one or both drugs may be either increased (**potentiating effect**) or decreased (**inhibiting effect**).

## DRUG INTERACTIONS



## Cont, Effects of drugs

**7. Drug misuse** is the improper use of common medications in ways that lead to acute and chronic toxicity.

**8. Drug abuse** is inappropriate intake of a substance, either continually or periodically. Drug abuse has two types' dependence and habituation.

هو استعمال المواد المخدّرة أو المؤثرة على العقل بطريقة خاطئة

**9. Drug dependence:** is a person's reliance on or need to take a drug or substance. The two types of dependence, physiologic and psychological. (الاعتماد الدوائي)

❖ **Physiologic dependence** is due to biochemical changes in body tissues, especially the nervous system. (الاعتماد الجسدي)

❖ **Psychological dependence** (الاعتماد النفسي) is emotional reliance on a drug to maintain a sense of well-being accompanied by feelings of need or cravings for that drug رغبة شديدة.

**١٠. Drug habituation:** (التعود الدوائي) denotes a mild form of psychological dependence. The individual develops the habit of taking the substance and feels better after taking it. هو شكل خفيف من الاعتماد النفسي



## Actions of drugs on the body (drug half-life)

- It means the time interval required for the body's elimination processes to reduce the concentration of the drug in the body by one-half life. هو الوقت اللي يحتاجه الجسم عشان يتخلص من ٥٠٪ من كمية الدواء الموجودة فيه



## **Actions of drugs on the body (drug half-life)**

Because the purpose of most drug therapy is to maintain a constant drug level in the body, repeated doses are required to maintain that level. For example, if a drug's half-life is 8 hours, then the amount of drug in the body is as follows:

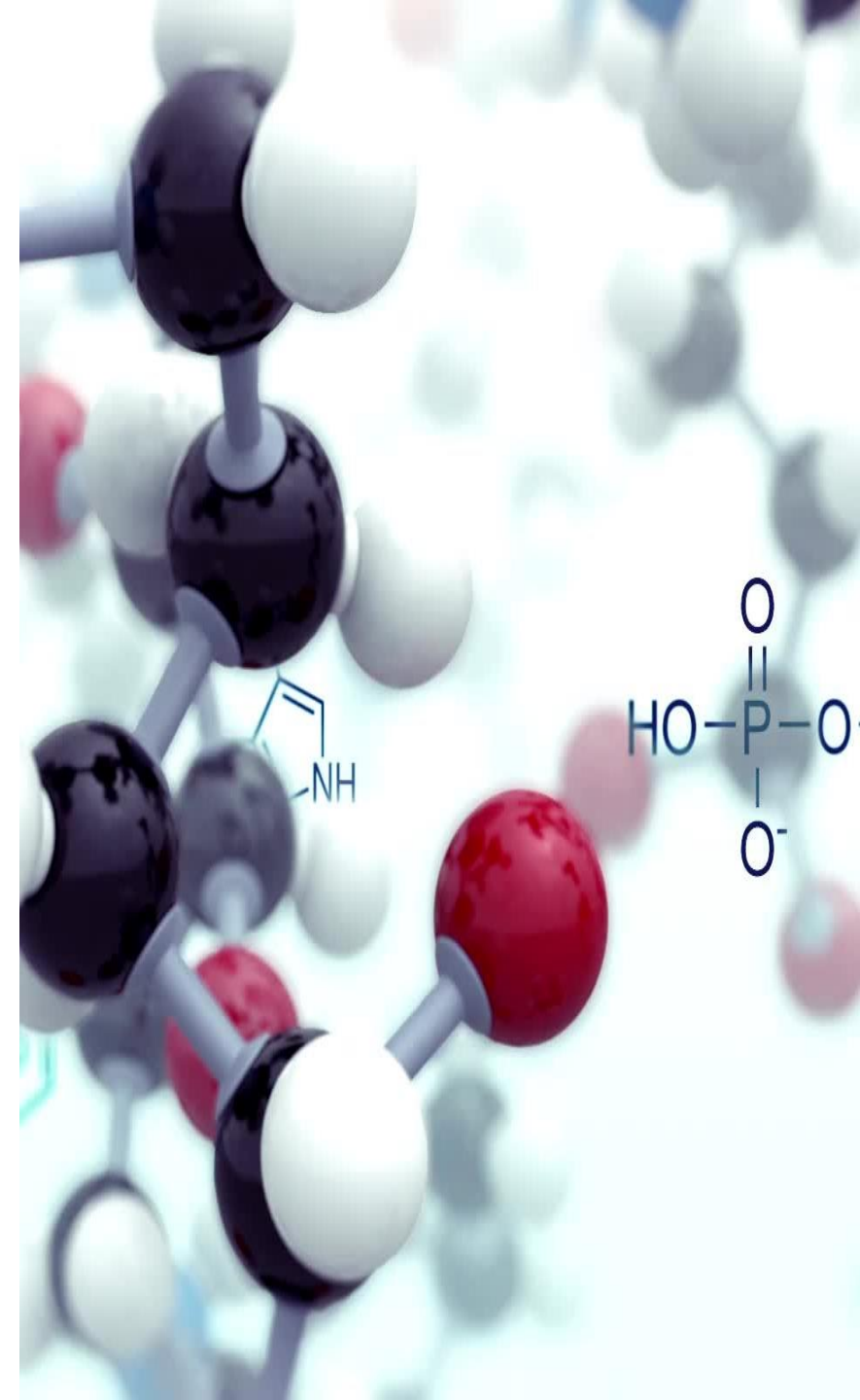
- ✓ Initially: 100%
- ✓ After 8 hours: 50%
- ✓ After 16 hours: 25%
- ✓ After 24 hours: 12.5%
- ✓ After 32 hours: 6.25%

# Pharmacodynamics

ماذا يفعل الدواء بالجسم؟

Is the mechanism of drug action and the relationships between drug concentration and responses in the body.

**A receptor:** is the drug's specific target, usually a protein located on the surface of a cell membrane or within the cell. As the drug binds to the receptor, it enhances or inhibits the normal cellular function. When a drug binds to its receptor, the pharmacologic effects are either **agonism** له تأثير يفعل or **antagonism** يمنع تفعيل ليس له تأثير.



# Pharmacodynamics

- **An agonist** is a drug produces the same type of response as the physiological or endogenous substance. **For example**, epinephrine like drugs acts on the heart to increase the heart rate.
- **An antagonist** is a drug that inhibits cell function by occupying receptor sites. The antagonist prevents natural body substances or other drugs from activating the functions of the cell by occupying the receptor sites.
- **For example**, naloxone is an opioid antagonist used as an antidote for respiratory depression caused by an opioid drug (e.g., morphine). This drug competes with opioid receptor sites in the brain and thereby prevents the opioid from binding to its receptors. By blocking the effect of the opioid, respiratory depression is reversed.

# Pharmacokinetics هي دراسة ما يفعله الجسم بالدواء بعد

دخوله الجسم

- Is the study of the **absorption**, **distribution**, **biotransformation**, and **excretion of drugs**.

**1. Absorption:** is the process by which a drug passes into blood stream



# Factors affecting the rate of drug absorption according to administration routes:

## 1. Oral route:

1. **Food** in the stomach can **delay the dissolution** and absorption of drugs.
2. **Food** can also **combine with molecules** of certain drugs and inhibiting their absorption.
3. **Acid medium** in the stomach affects absorption of drug.
4. **Some drugs do not dissolve** or have limited ability to dissolve in the gastrointestinal fluid.
5. **Some drugs are absorbed by tissues** before they reached to the stomach. For example, nitroglycerine is administered under the tongue, where it is absorbed into blood vessels that carry it directly to the heart, the intended site of action. If it swallowed, it will be destroyed by the liver. This is known as the **first- pass effect**, before reaching the intended site .

## **Cont, Factors affecting the rate of drug absorption according to administration routes:**

**2. Intravenous route:** is the route of choice for rapid action.

**3. Intramuscular route:** is the next most rapid route.

**4. Subcutaneous route:** Absorption is slower. The rate of absorption can be accelerated by application of heat (which increase the blood flow to the area) and can be slowed by application of cold.

## Cont, Pharmacokinetics

**5. Rectum route:** Absorption of drugs from the bloodstream tends to be unpredictable. Therefore, this route is normally used when other routes are unavailable or when the intended action is localized to the rectum or sigmoid colon.

**2. Distribution** is the transportation of a drug from its site of **absorption** to **its site of action**.

When a drug enters bloodstream, it is carried to the most vascular organs (liver, kidneys, and brain). Body areas with lower blood supply (skin and muscles) receive the drug later.

Chemical and physical Properties of drug determine the area of the body to which the drug will be attracted. For example, fat- soluble drugs will accumulate in fatty tissues. حركة الدواء في

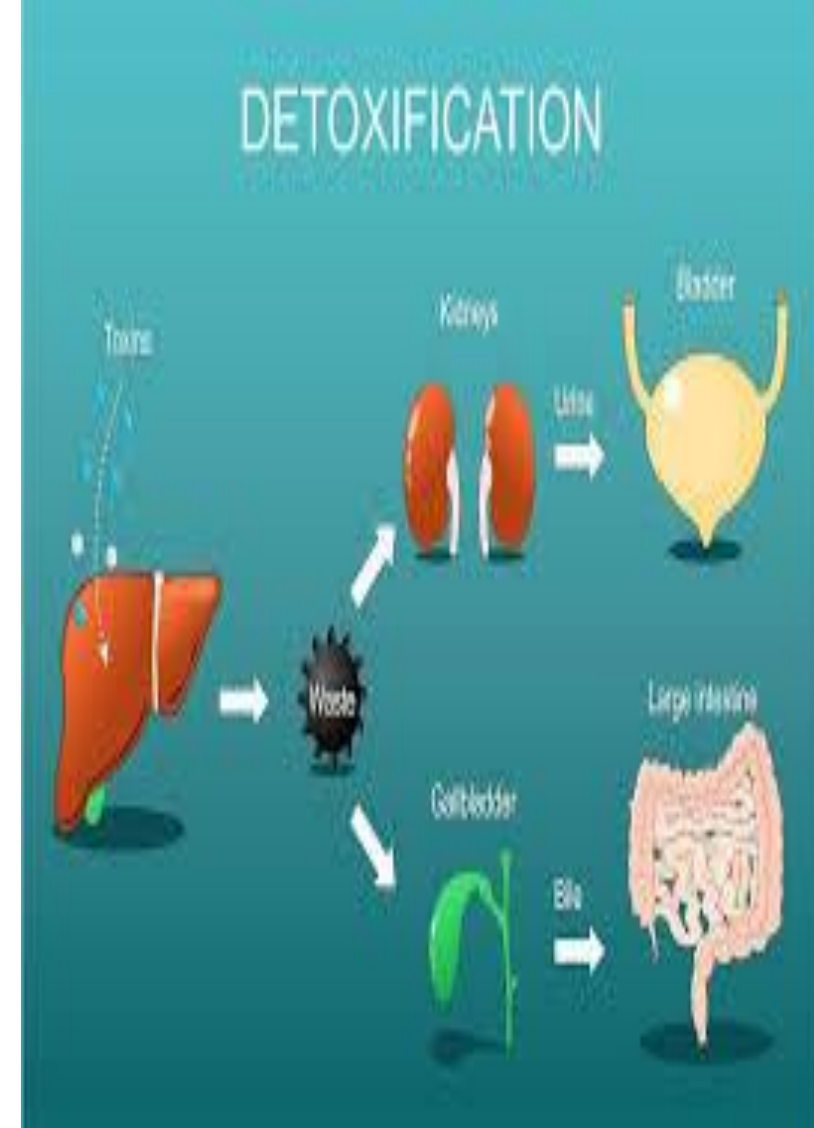
الجسم بعد دخوله الدم حتى يصل إلى مكان تأثيره.



### 3. Biotransformation عملية كيميائية يقوم بها الجسم لتحويل المواد الغريبة أو الأدوية إلى أشكال أسهل في الإخراج (التحول الحيوى) metabolism of drug

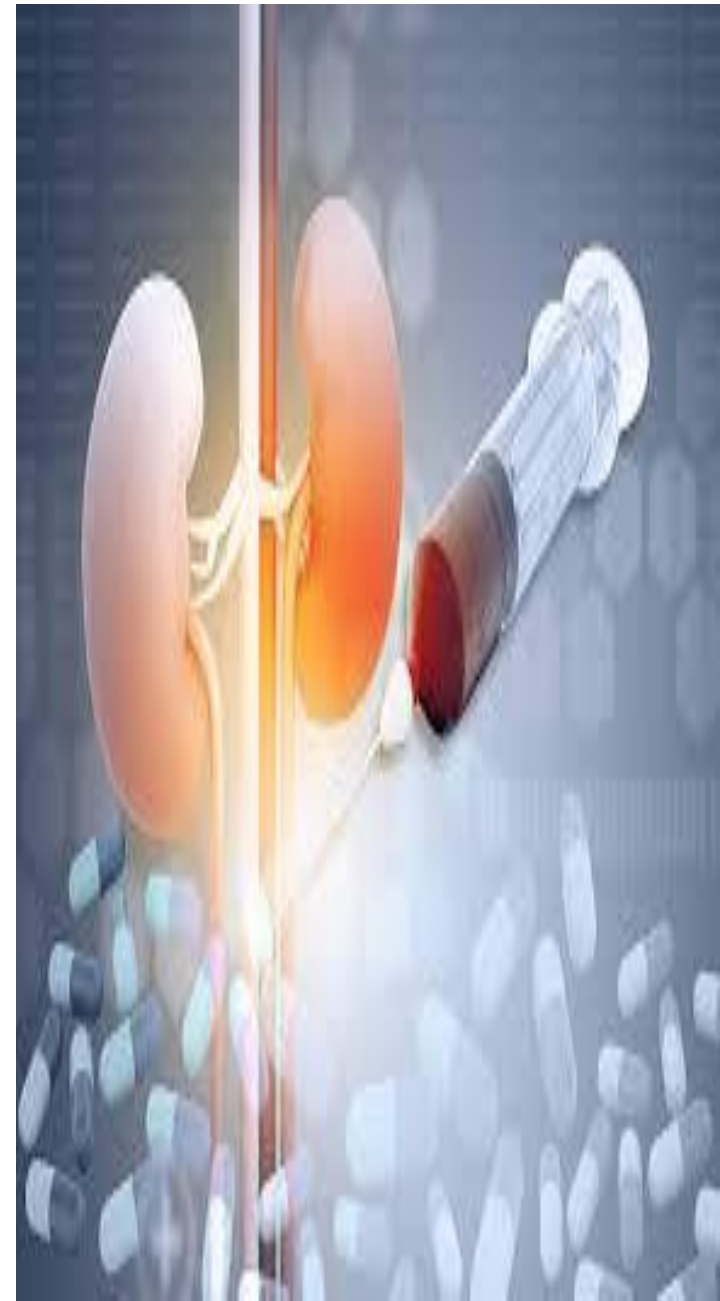
Also called detoxification اخراج السموم or metabolism, is a process by which a drug is converted to a less active form. Most biotransformation takes place in the liver, where many drug metabolizing enzymes in the cells detoxify the drugs. The products are called metabolites. Biotransformation may be altered if a person is very young, is older, or has an unhealthy liver.

عندما يدخل دواء أو مادة سامة إلى الجسم، يقوم الجسم  
تعديلها كيميائياً  
جعلها أكثر ذوباناً في الماء  
تسهيل التخلص منها عبر البول أو الصفراء

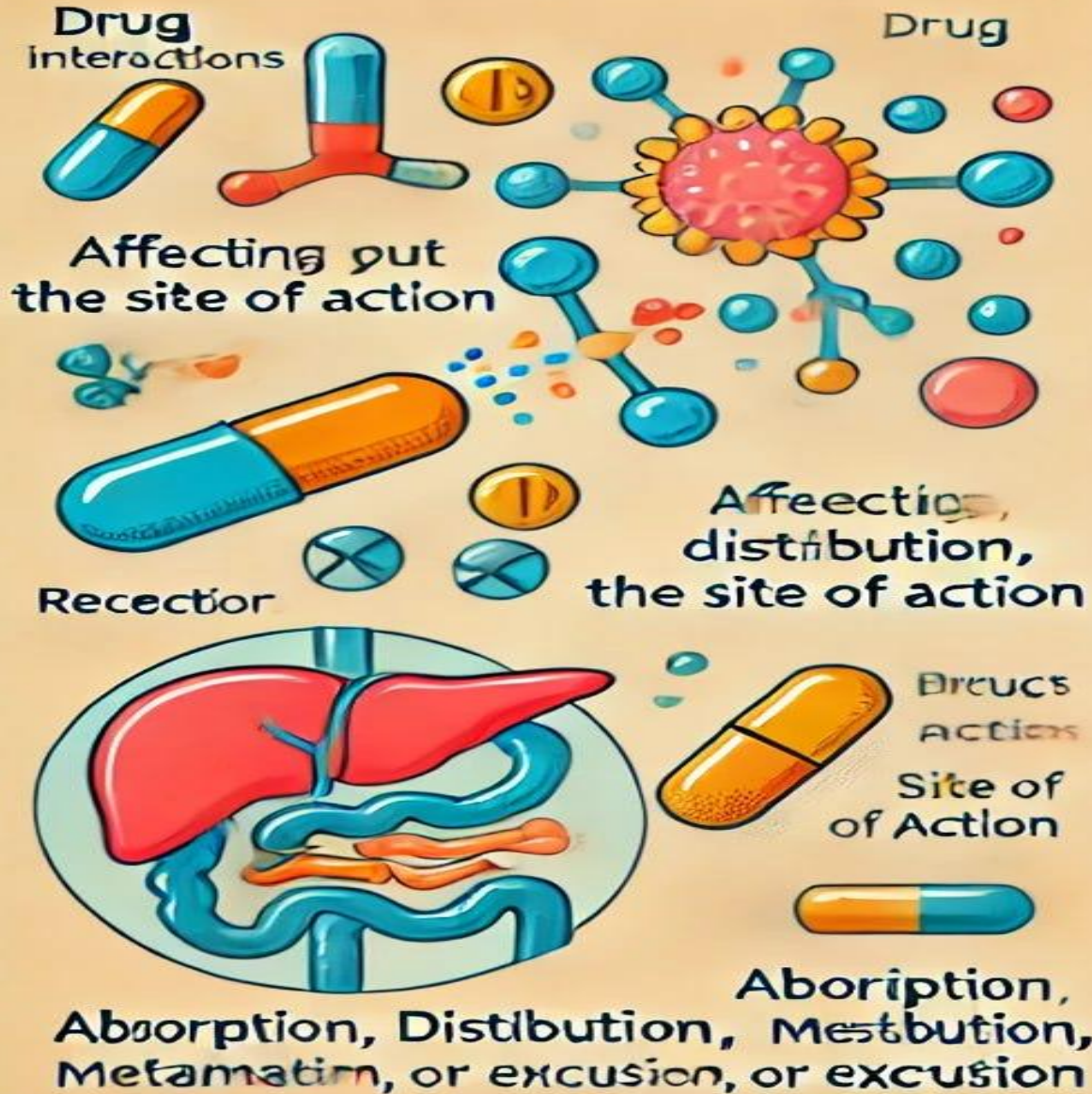




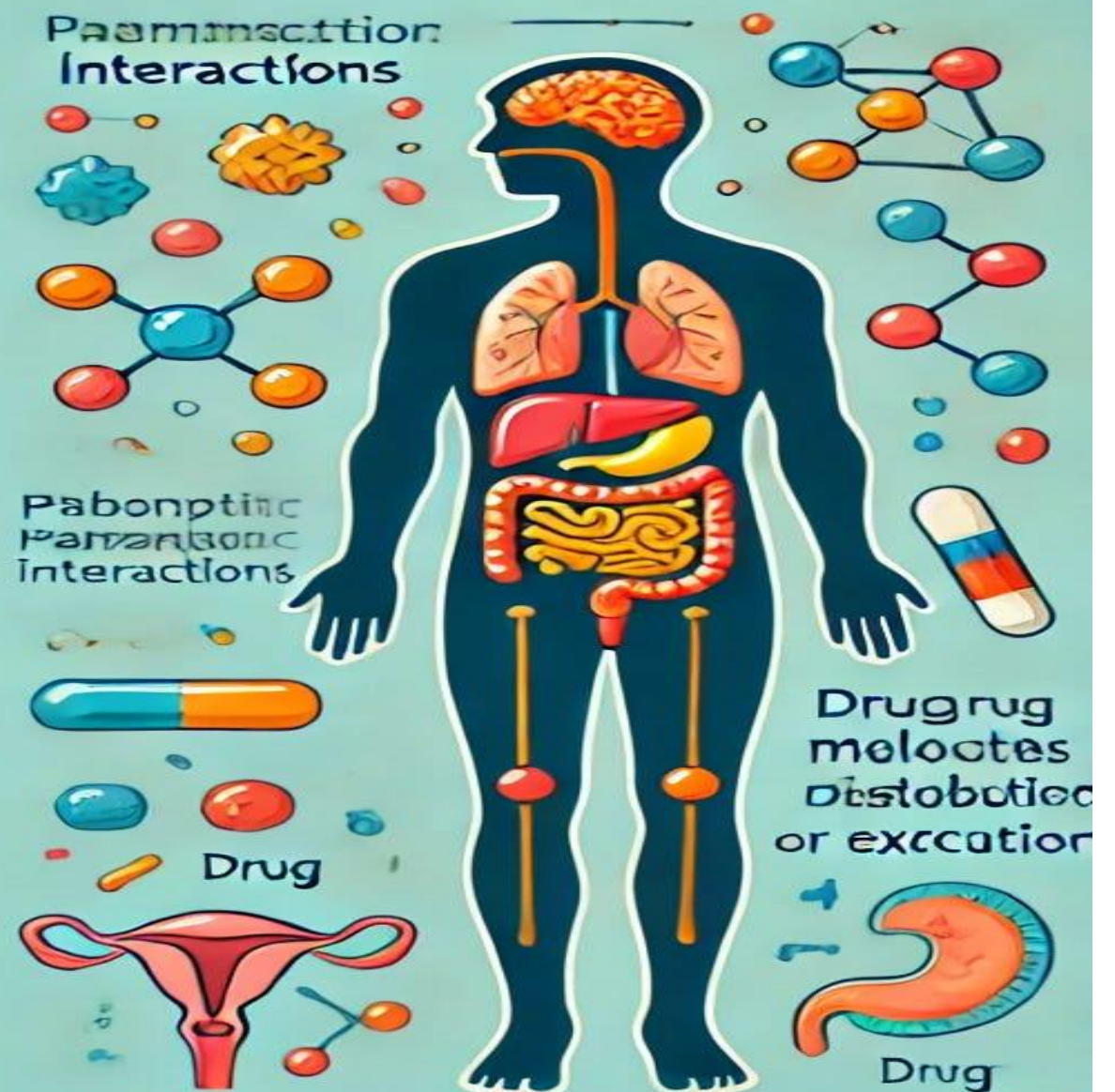
**4. Excretion** is the process by which metabolites and drugs are **eliminated from the body**. Most metabolites are eliminated by the **kidneys in the urine**; however, some are **excreted in feces, breath, perspiration, saliva, and breast milk**. The efficiency of the kidneys to excrete drugs **diminishes with age. Older people** may require smaller doses of a drug because the drug and its metabolites may accumulate in the body.



# Pharmacodynamic Interactions



# Pharmokinotic Interactions







# Factors affecting medication action

## 1. Developmental factors

- **During pregnancy** women must be very careful about taking medications. Drugs taken during pregnancy pose a risk throughout pregnancy, but pose the highest risk during the first trimester, due to the formation of vital organs and functions of the fetus during this time.
- **Infants** usually require small dosages because of their body size and immaturity of their organs, especially the liver and kidneys.



# Factors affecting medication action

- **Older adults** have different responses to medications due to physiologic changes that accompany aging. These changes include
  - **Decreased liver and kidney function**, which can result in accumulation of the drug in the body.
  - Older adults often experience **decreased gastric motility and decreased gastric acid production. Increased adipose tissue and decreased total body fluid proportionate** to the body mass can increase the possibility of drug toxicity.



# Cont, Factors affecting medication action

**2. Gender:** Differences in the way **men** and **women** respond to drugs are chiefly related to the distribution of **body fat and fluid** and **hormonal differences**.

**3. Diet:** Nutrients can affect the action of medication. For example, **vitamin K** found in green leafy vegetables can counteract the **effect of anticoagulant** such as warfarin.

## Cont, Factors affecting medication action

**4. Environment:** Temperature may also affect drug activity.

- ❖ When environmental temperature is **high the peripheral blood vessels dilate**, thus intensifying the action of vasodilators.
- ❖ In contrast, a cold environment and the consequent **vasoconstriction inhibit the action** of vasodilators but **enhance the action of vasoconstrictors**.
- ❖ A client who takes **a sedative or analgesic in a busy**, noisy environment may not benefit as fully as if the environment were quiet and peaceful.

## Factors affecting medication action

**5. Psychological factors:** A client's expectations about what a drug can do can affect the response to medication. For example, a client who believes that codeine is ineffective as an analgesic may experience no relief from pain after it is given.

**6. Illness and disease:** Illness and disease can also affect the action of drugs. For example, aspirin can reduce the body temperature of a feverish client but has no effect on the body temperature of a client without fever. Drug action is altered in clients with circulatory, liver, or kidney dysfunction.



## Cont, Factors affecting medication action

**7. Time of administration:** The time of administration of oral medications affects the relative speed with which they act.

❑ Orally administered medications are also **absorbed more quickly if the stomach is empty**. Thus, oral medications **taken 2 hours before meals**, act faster than those taken **after meals**.

❑ However, some medications, for example iron preparations, **irritate the gastrointestinal tract** and are given **after a meal**.

# 8. Routes of Administration

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## ➤ 1. Oral Routes

- The oral route is the **easiest** and **the most commonly used route**. Medications are given by mouth and swallowed with fluid.
- Oral medications have **a slower onset of action** and a more prolonged effect than parenteral medications. Patients generally prefer the oral route.



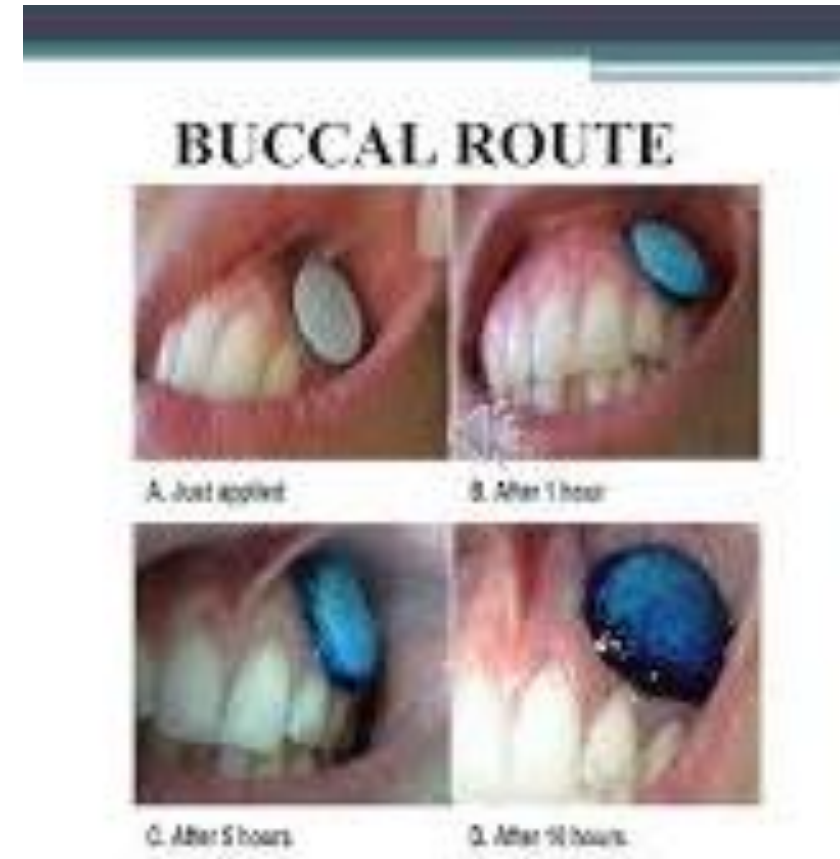
## ➤ 2. Sublingual Administration

- ❖ Some medications are readily absorbed after being placed under the tongue to dissolve.
- ❖ A medication given by the **sublingual** route should not be swallowed because the medication does not have the desired effect.
- ❖ Nurses often give nitroglycerin by the sublingual route.  
Tell the patient **not to drink anything until the medication is completely dissolved.**



# 3. Buccal Administration

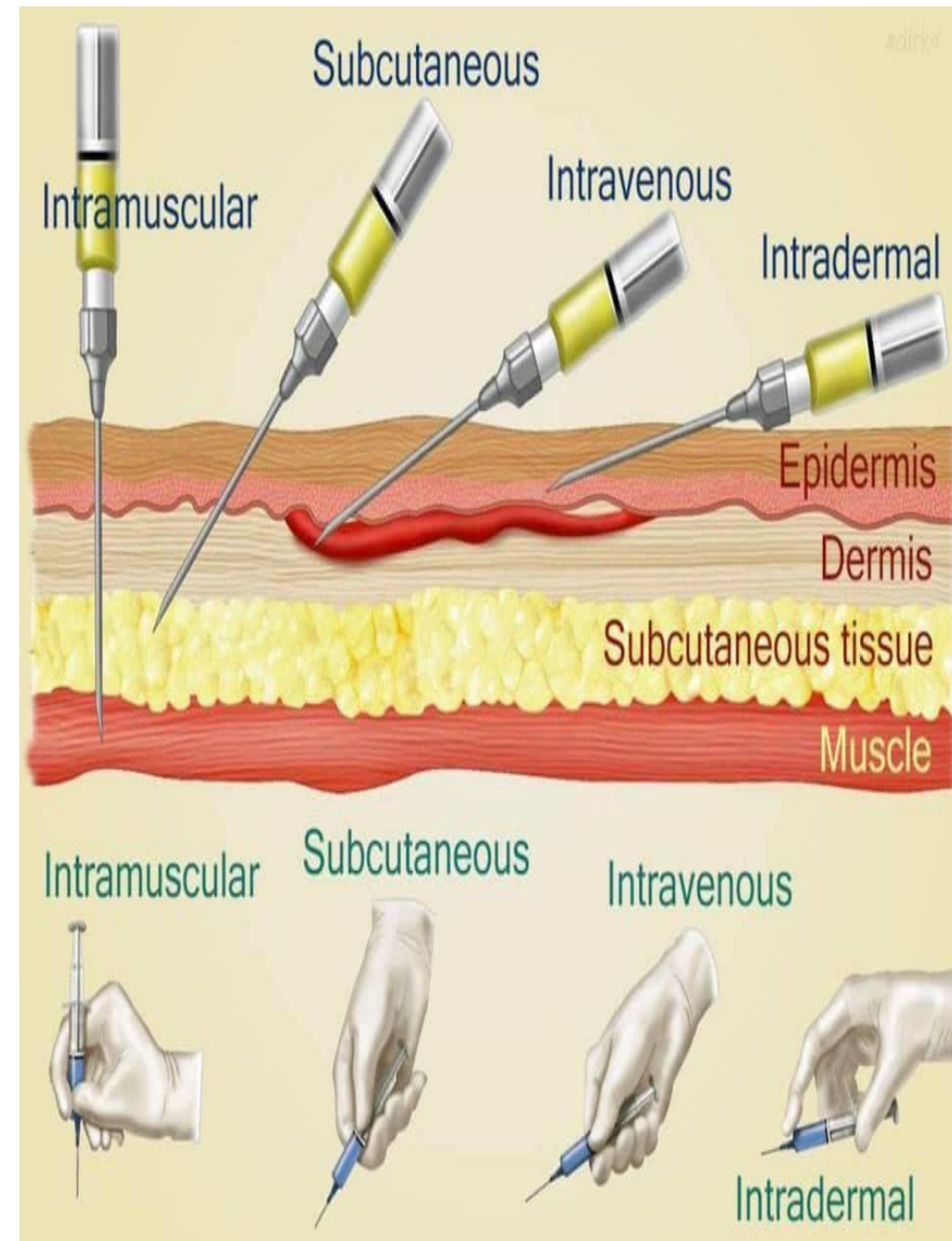
- ❖ Administration of a medication by the **buccal route** involves placing the solid medication in the mouth against the mucous membranes of the cheek until it dissolves.
- ❖ Teach patients to **alternate cheeks with each subsequent dose to avoid mucosal irritation.**
- ❖ **Warn patients not to chew or swallow the medication or to take any liquids with it.**
- ❖ A buccal medication **acts locally on the mucosa or systemically as it is swallowed in a person's saliva.**



## 4. Parenteral Routes

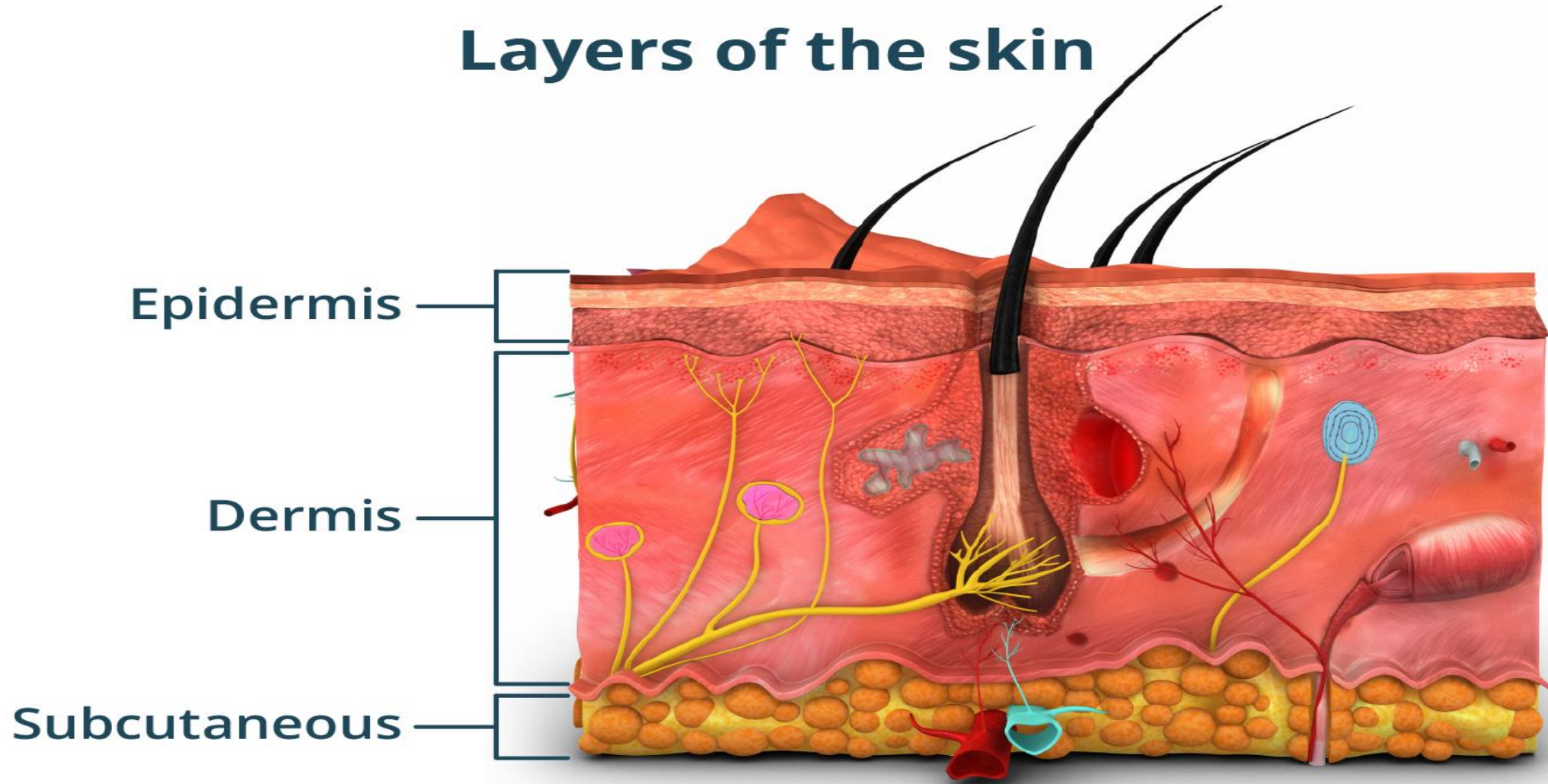
- Parenteral administration involves injecting a medication into body tissues. The following are the **four major sites** of injection:

- Intradermal (ID):** Injection into the dermis just under the epidermis
- Subcutaneous:** Injection into tissues just below the dermis of the skin
- Intramuscular (IM):** Injection into a muscle
- Intravenous (IV):** Injection into a vein





# Layers of the skin



## 5. Topical Administration

- ✓ Medications applied to the skin and mucous membranes generally have **local effects**.
- ✓ Application of topical medications to the skin by painting or spreading the medication over an area, applying moist dressings, soaking body parts in a solution, or giving medicated baths.
- ✓ **Systemic effects** often occur if a patient's skin is thin or broken down, the medication concentration is high, or contact with the skin is prolonged.





## 6. Inhalation Route

- The deeper passages of the respiratory tract provide a large surface area for medication absorption.



- Nurses administer inhaled medications through the

- **Nasal**
- **Oral passages**
- **Endotracheal tubes**
- **Tracheostomy tubes.**





# Types of medication orders

**1. A stat order** indicates that medication is to be given immediately and only once. حالا ولمرة واحدة

1. STAT orders are often written for **emergencies** when a patient's condition changes suddenly.

**For example:** Apresoline 10mg IV STAT



## Cont, Types of medication orders

**2. The single order** or one-time order is for medication to be given once at a specified time. This is common for **preoperative medications** or **medications given before diagnostic examinations**. **For example:** Ativan 1mg IV on call to magnetic resonance imaging (MRI)



## Cont, Types of medication orders

### 3. The Standing order يعطى الدواء بشكل منتظم قد يكون بدون تاريخ انتهاء

او لمدة محددة may or may not have a termination date. It may be

carried out until an order is written to cancel it or it may be carried

for specified days. **Examples** of standing orders: Tetracycline

500 mg PO q6h and Decadron 10 mg daily × 5 days



## Cont, Types of medication orders

⚡.A **prn** order or as needed order, permit the nurse to give a medication when, in nurses judgment, the client requires it.  
Example: Morphine sulfate 2mg IV q2h prn for incisional pain.





## Brief comparison between types of medication order

| النوع    | متى يُعطى؟  | عدد المرات          |
|----------|-------------|---------------------|
| STAT     | فورًا       | مرة واحدة           |
| Single   | في وقت محدد | مرة واحدة           |
| Standing | بشكل منتظم  | مستمر أو لمدة محددة |
| PRN      | عند الحاجة  | حسب الحاجة          |

# Essential parts of a drug order

- ♦ Full name of the client
- ♦ Date and time the order is written
- ♦ Name of the drug to be administered
- ♦ Dosage of the drug
- ♦ Frequency of administration
- ♦ Route of administration
- ♦ Signature of the person writing the order.

**Example :** demerol 100 mg IM q4h

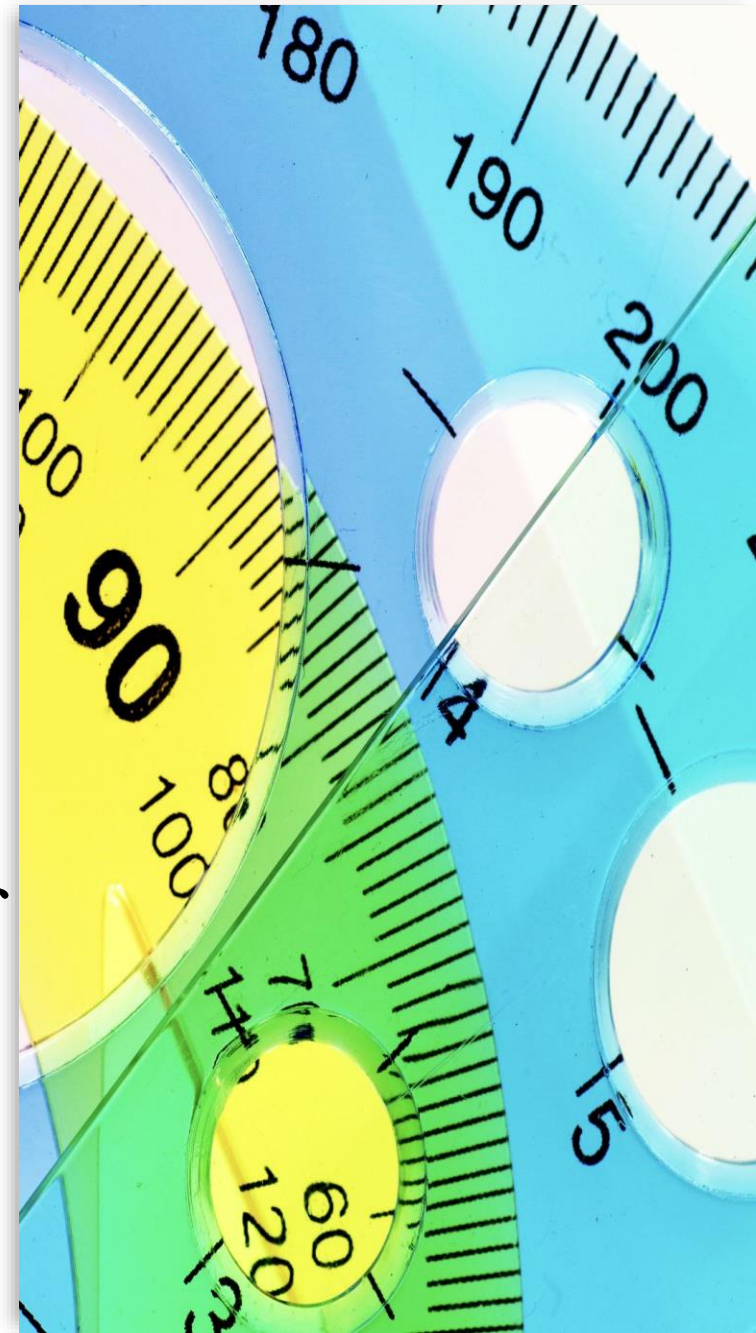


# Systems of Medication Measurement

- **Metric System** (النظام المتري) وحدات الوزن والحجم فقط
  - For medication calculations only use the volume and weight units. In the metric system use lowercase or capital letters to designate basic units:
- **Capital letters**
  - Gram= g or Gm
  - Liter = l or L
- **lowercase letters** for abbreviations for other units:
  - Milligram =mg
  - Milliliter =mL

# Household Measurements (الوحدات المنزلية)

- Household units of measure are familiar and convenience to most people
- But have **disadvantage** that include **inaccuracy** as household **الادوات** utensils (teaspoons and cups) vary in size. For calculation medications accurately, **غير دقيقة لان الحجم يختلف من منزل لآخر**
- The nurse need to know common equivalents of metric and household units.



**Table (3): Equivalents of Household Measurement**

| Metric | Household            |
|--------|----------------------|
| 1 Ml   | 15 drops (gtt)       |
| 5 Ml   | 1 teaspoon (tsp)     |
| 15 mL  | 1 tablespoon (tbsp)  |
| 30 mL  | 2 tablespoons (tbsp) |
| 240 mL | 1 cup (c)            |





